

Lecture 8a

EE-215 Electronic Devices and Circuits

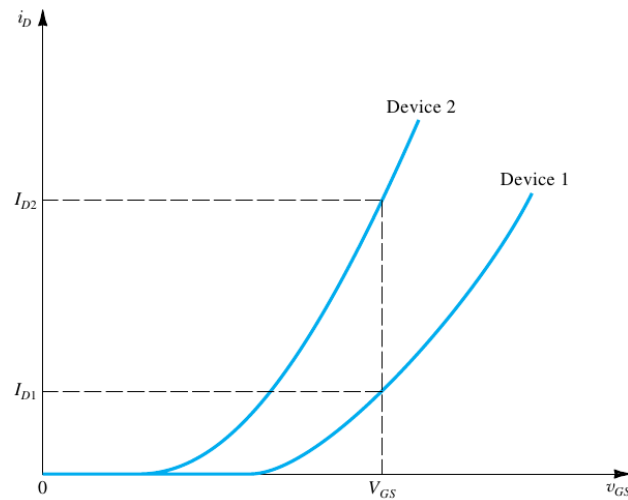
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Biasing in MOS Amplifier Circuits

- we have already determined
 - In the design of a MOS amplifier circuit,
 - it is essential to establish an appropriate dc operating point (or Q point or bias point) for the MOSFET
 - This step is known as biasing or bias design
 - An appropriate bias point or dc operating point is characterized by
 - a stable and predictable dc drain current I_D and
 - by a dc drain-to-source voltage V_{DS} that ensures operation in the saturation region for all expected input signal levels

Biasing by Fixing V_{GS}

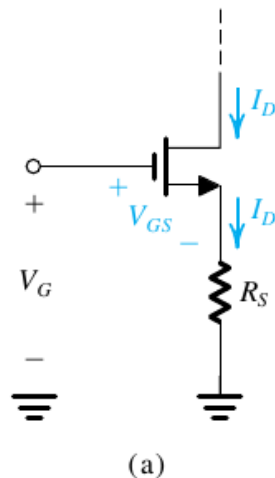
- The simplest way to bias a MOSFET is to fix its gate-to-source voltage, V_{GS}
 - to the value that is required for the desired I_D
 - This voltage can be easily derived from the power-supply V_{DD}
 - by using an appropriate voltage divider
 - however, this way of biasing a MOSFET is not a good approach
 - because practically there are always some variations in V_t , C_{ox} among the devices of same type
 - As $I_D = \frac{1}{2}\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_t)^2$
 - if V_t , C_{ox} of two devices is different
 - $\Rightarrow I_D$ will be different for the same value of V_{GS}
- also, V_t and μ_n are temperature dependent, and if we fix the value of V_{GS} ,
 - the drain current I_D will become very temperature dependent
 - thus for the fixed value of V_{GS} , the resultant spread
 - in the value of drain current can be substantial
 - ($\because$ of change in V_t , C_{ox} , μ_n among the devices of same type)



■ **Figure 5.51** The use of fixed bias (constant V_{GS}) can result in a large variability in the value of I_D . Devices 1 and 2 represent extremes among units of the same type.

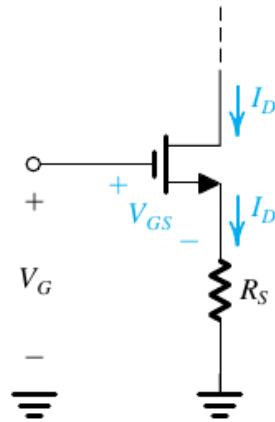
Biasing by Fixing V_G and Connecting a Resistance in the Source

- An effective biasing technique for discrete MOSFET circuits,
 - consists of fixing the dc voltage at the gate V_G
 - and connecting a resistance in the source lead
 - By KVL , $V_G = V_{GS} + I_D R_S \Rightarrow I_D = \frac{V_G - V_{GS}}{R_S}$
 - from this relation, if V_G is much greater than V_{GS}
 - $\Rightarrow I_D$ will be mostly determined by V_G and R_S
 - even when V_G is not much larger than V_{GS} , resistor R_S
 - provides a negative feedback which acts to stabilize the value of the bias current I_D



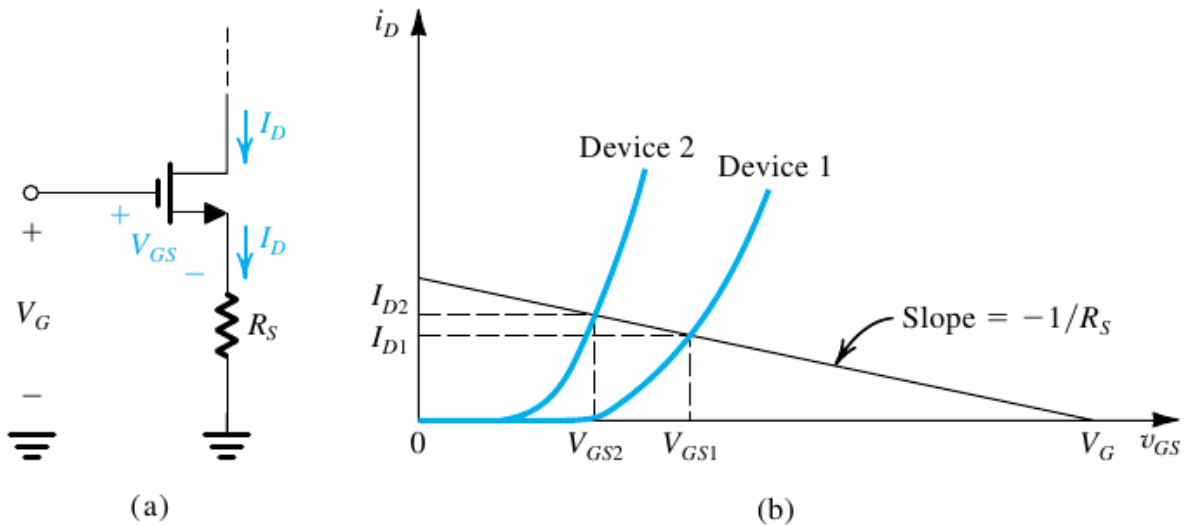
- $V_G = V_{GS} + I_D R_S \Rightarrow I_D = \frac{V_G - V_{GS}}{R_S}$
 - even when V_G is not much larger than V_{GS} , resistor R_S

- provides a negative feedback which acts to stabilize the value of the bias current I_D
- negative feedback here, means that if I_D increases for any reason,
 - the equation ($V_G = V_{GS} + I_D R_S$) indicates that since V_G is a constant,
 - V_{GS} will have to decrease. this in turn results in a decrease in I_D



(a)

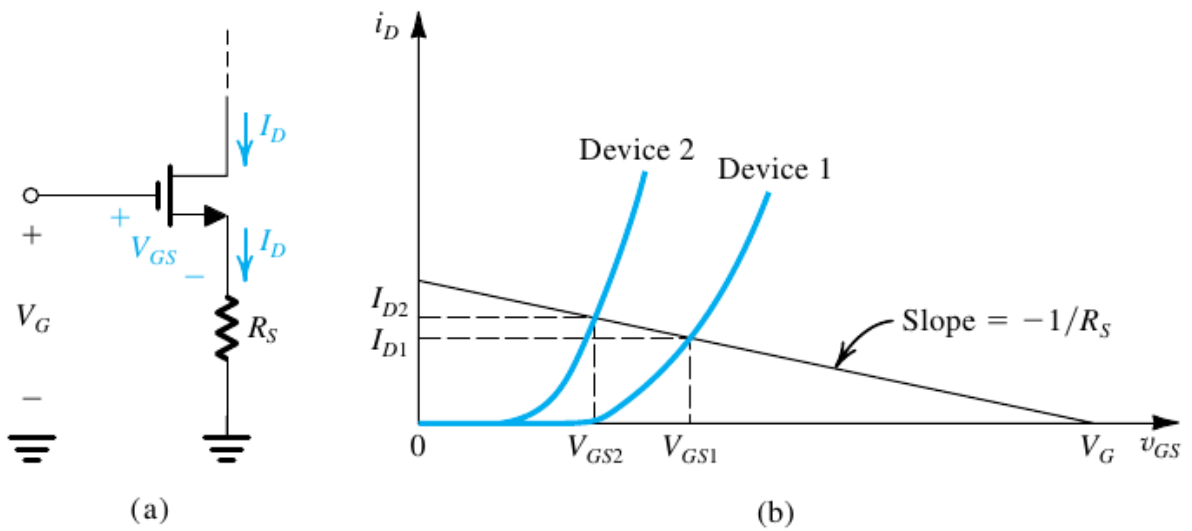
- Thus the purpose of R_S is to keep I_D as constant as possible



(a)

(b)

- here $i_D - v_{GS}$ characteristics for two devices that represent the extremes of a batch of MOSFETs, are shown
- The straight line, (superimposed on the device characteristics), represents the constraint imposed by the bias circuit
 - i.e. $V_G = V_{GS} + I_D R_S$



- The intersection of this straight line ($V_G = V_{GS} + I_D R_S$) with the $i_D - v_{GS}$ characteristic curve
 - provides the coordinates (I_D, V_{GS}) of the bias point.

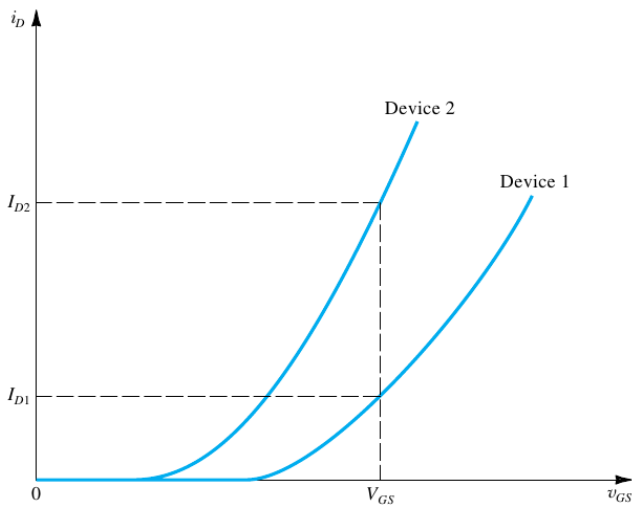
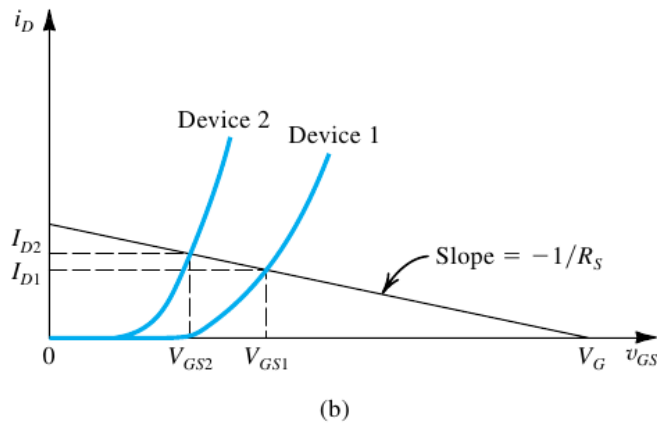
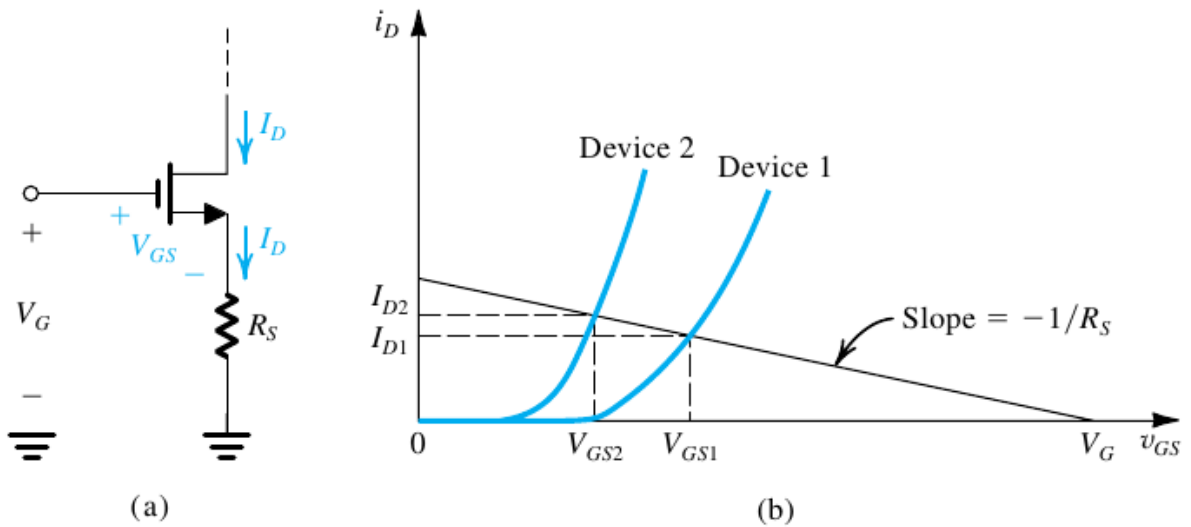


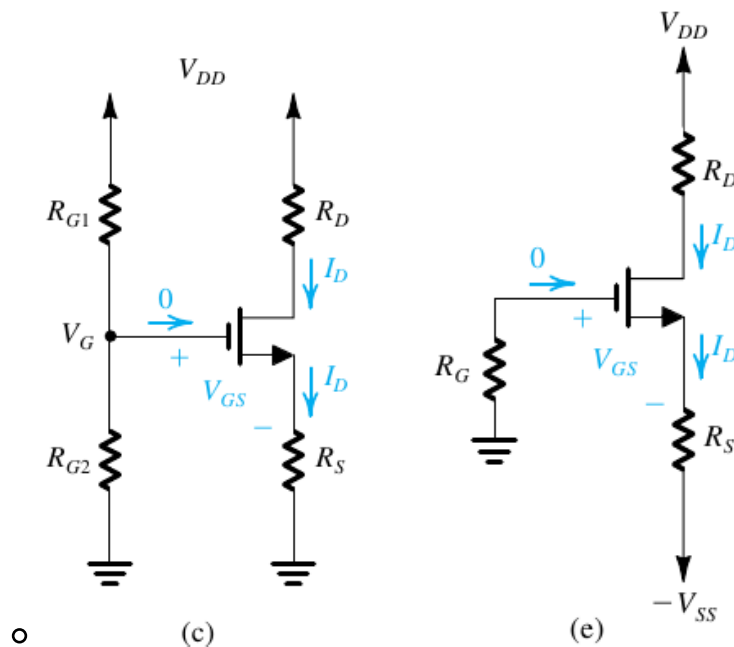
Figure 5.51 The use of fixed bias (constant V_{GS}) can result in a large variability in the value of I_D . Devices 1 and 2 represent extremes among units of the same type.



- note that compared to the case of fixed V_{GS} (left fig),
 - here (right fig) the variation in I_D is much smaller



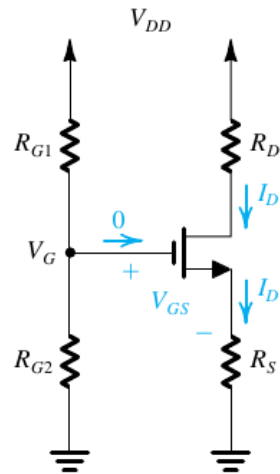
- - Note that the variation in I_D decreases as V_G and R_S are made larger
 - (i.e. the bias line is less steep when V_G and R_S are larger)
- Two possible practical discrete implementations of this bias scheme are shown in fig



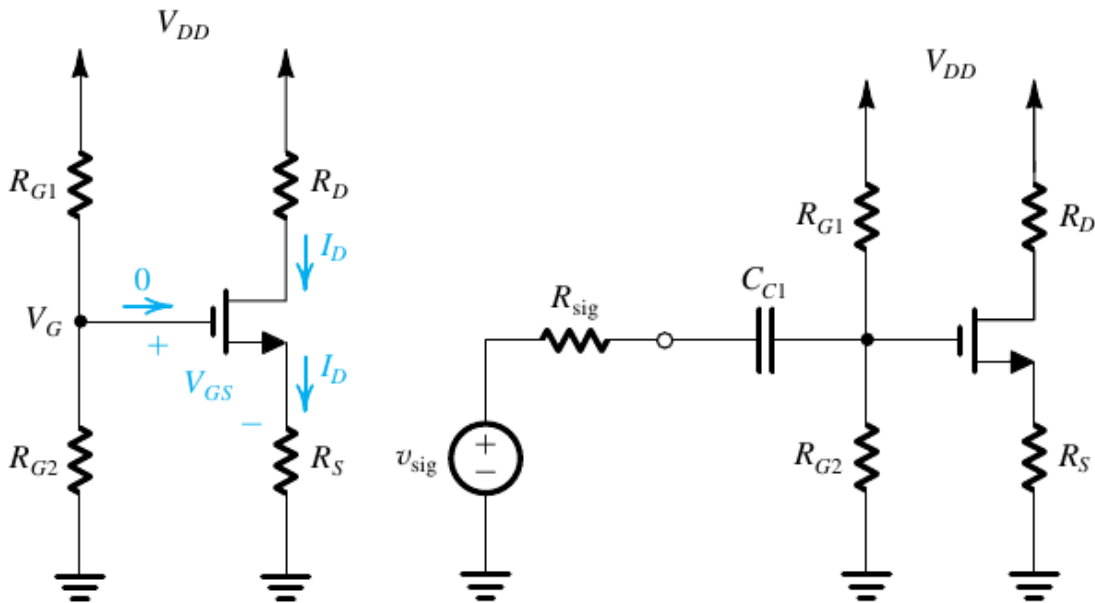
- here just one power supply V_{DD} is required
 - V_G is derived using a voltage divider (R_{G1}, R_{G2})
 - As $I_G = 0$, R_{G1} and R_{G2} can be selected to be very large (usually in Mega ohms range)
 - this allows the MOSFET to present a large R_{in} to the

signal source

- that can be connected to the gate through a coupling capacitor



(c)

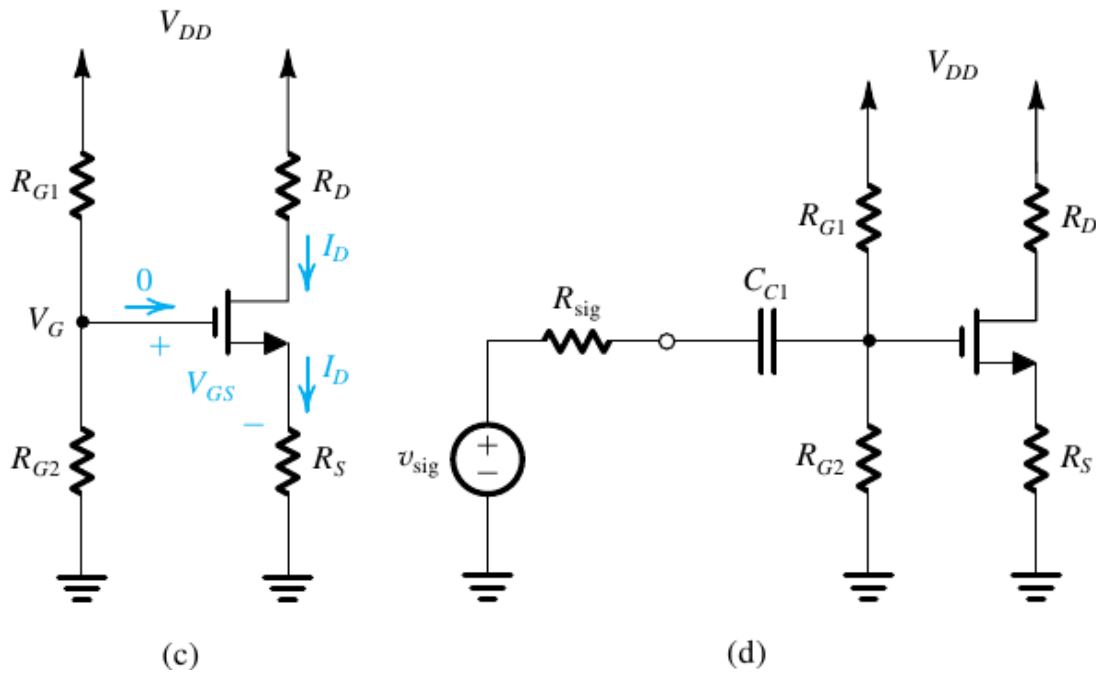


(c)

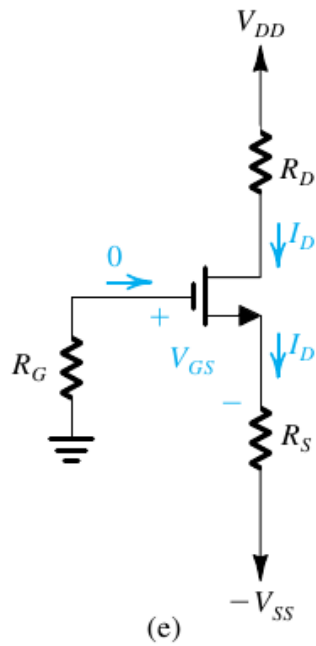
(d)

○ here capacitor C_{C1} blocks dc and thus enables us to couple

- the signal v_{sig} to the amplifier input without disturbing the MOSFET bias point

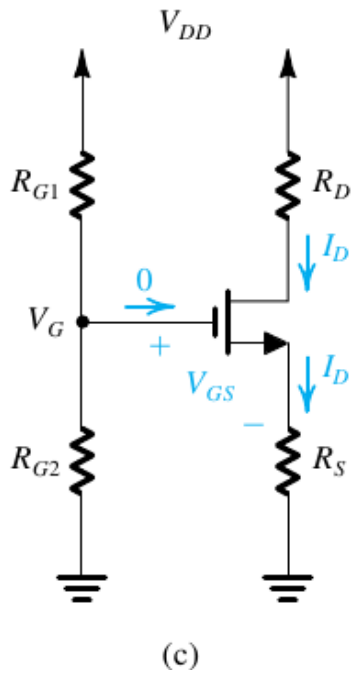


- - The value of C_{C1} should be selected large enough to
 - approximate a short circuit at all signal frequencies of interest
- if two power supplies are available, a relatively simpler biasing scheme can be used
 - as gate current is zero $\Rightarrow V_G = 0$
 - by KVL, $V_G = V_{GS} + I_D R_S + (-V_{SS})$
 - $0 = V_{GS} + I_D R_S - V_{SS} \Rightarrow V_{SS} = V_{GS} + I_D R_S$
 - Note that this relation is of the same form as the relation for the previous circuit ($V_G = V_{GS} + I_D R_S$), if V_G is replaced by V_{SS}
 - Note that R_G establishes a dc ground at the gate
 - and presents a high input resistance to a signal source
 - which may be connected to the gate through a coupling capacitor



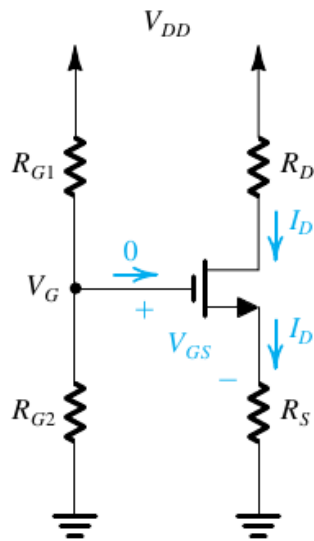
Example 5.12

- It is required to design the circuit of Fig. 5.52(c) to establish a dc drain current $I_D = 0.5\text{mA}$. The MOSFET is specified to have $V_t = 1\text{V}$ and $k_n'(W/L) = 1\text{mA/V}^2$. For simplicity, neglect the channel-length modulation effect (i.e., assume $\lambda = 0$). Use a power-supply $V_{DD} = 15\text{V}$. Calculate the percentage change in the value of I_D obtained when the MOSFET is replaced with another unit having the same $k_n'(W/L)$ but $V_t = 1.5\text{V}$.



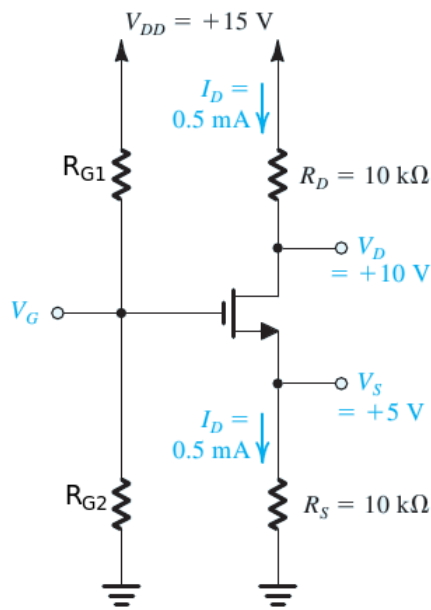
Solution:

- As a rule of thumb, this biasing circuit can be designed by
 - selecting R_D and R_S to provide one-third of the V_{DD}
 - as a voltage drop across each of the
 - R_D , the transistor (i.e. V_{DS}) and R_S
 - i.e. $I_D R_D = V_{DS} = I_D R_S = \frac{1}{3} V_{DD} = 5V$
 - $\Rightarrow V_D = 10V$, $V_S = 5V$
 - here 1st two conditions ($I_D R_D = V_{DS} = \frac{1}{3} V_{DD}$) will ensure enough signal swing
 - 3rd condition ($I_D R_S = \frac{1}{3} V_{DD}$) will ensure bias point stability

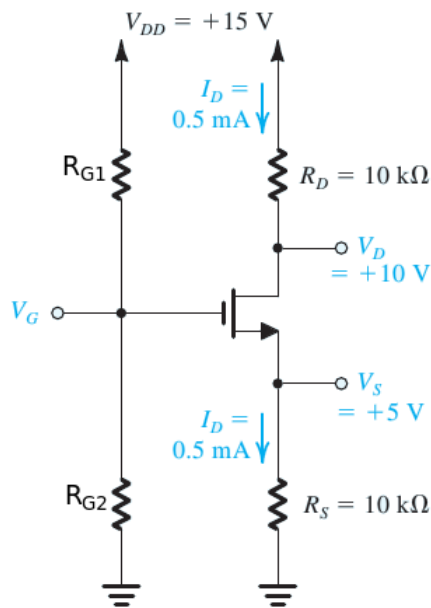


(c)

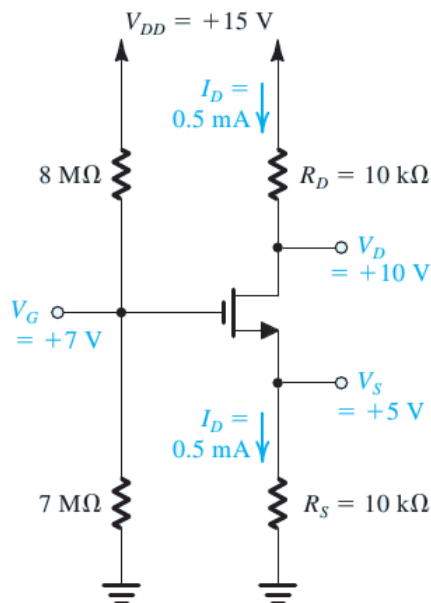
- here $I_D = 0.5mA$, $V_{DD} = 15V$, $V_t = 1V$, $k_n'(W/L) = 1mA/V^2$, $\lambda = 0$
 - as $I_D R_D = I_D R_S = 5V$, $I_D = 0.5mA$
 - $\Rightarrow R_D = R_S = \frac{5}{I_D} = \frac{5}{0.5mA} = 10k\Omega$
 - to determine V_{GS}
 - $I_D = 0.5mA = \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2$
 - $0.5mA = \frac{1}{2} (1mA/V^2) (V_{GS} - 1)^2$
 - $1 = (V_{GS} - 1)^2 \Rightarrow (V_{GS} - 1) = \sqrt{1} = \pm 1$
 - $(V_{GS} - 1) = 1 \because V_{OV}$ cannot be -ve if $I_D = 0.5mA$
 - $V_{GS} = 2V$



- here $I_D = 0.5\text{mA}$, $V_{DD} = 15\text{V}$, $V_t = 1\text{V}$, $k_n'(W/L) = 1\text{mA/V}^2$, $\lambda = 0$
 - $V_{GS} = 2\text{V}$
 - $\Rightarrow V_G - V_S = 2\text{V} = V_G - 5\text{V} \therefore V_S = 5\text{V}$
 - $V_G - 5 = 2 \Rightarrow V_G = 7\text{V}$
 - now to determine R_{G1} and R_{G2}
 - for $I_{RG1} = I_{RG2} = 1\mu\text{A}$
 - $\Rightarrow R_{G2} = \frac{V_G}{I_{RG2}} = \frac{7}{1\mu} = 7\text{M}\Omega$
 - also by ohms law
 - $V_{DD} = I_{RG1}(R_{G1} + R_{G2})$
 - $15 = 1\mu(R_{G1} + 7\text{M})$
 - $R_{G1} + 7\text{M} = 15\text{M} \Rightarrow R_{G1} = 8\text{M}\Omega$



- here $I_D = 0.5\text{ mA}$, $V_{DD} = 15\text{ V}$, $V_t = 1\text{ V}$, $k_n'(W/L) = 1\text{ mA/V}^2$, $\lambda = 0$
 - Note that the dc voltage at the drain is $+10\text{ V}$ and $V_{DD} = +15\text{ V}$
 - $\Rightarrow$ maximum +ve signal swing $= 15 - 10 = 5\text{ V}$
 - for maximum -ve signal swing
 - $V_{DS} \geq V_{GS} - V_t = 2 - 1 = 1\text{ V}$
 - $V_{DS} \geq 1\text{ V}$
 - here $V_{DS} = 5\text{ V}$
 - $\Rightarrow$ maximum -ve signal swing is -4 V (i.e. down to $V_{GS} - V_t = 1\text{ V}$)



- if now the MOSFET is replaced with another having $V_t = 1.5\text{ V}$,

$$k_n'(W/L) = 1\text{mA/V}^2, \lambda = 0$$

◦ the new I_D is

$$\blacksquare I_D = \frac{1}{2}k_n'(W/L)(V_{GS} - V_t)^2$$

$$\blacksquare I_D = \frac{1}{2}(1\text{m})(V_{GS} - 1.5)^2 = 0.5\text{m}(V_{GS} - 1.5)^2$$

$$\blacksquare \text{As } V_G = V_{GS} + I_D R_S$$

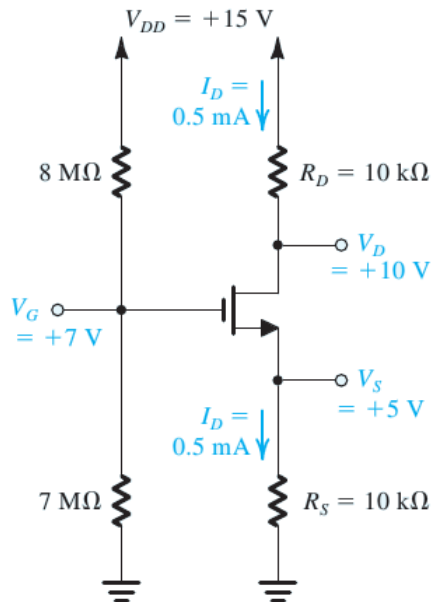
$$\blacksquare = \Rightarrow 7 = V_{GS} + I_D(10\text{k})$$

$$\blacksquare \text{or } I_D = \frac{7 - V_{GS}}{10\text{k}}$$

◦ eliminating I_D from the above two equations

$$\blacksquare I_D = 0.5\text{m}(V_{GS} - 1.5)^2 = \frac{7 - V_{GS}}{10\text{k}}$$

$$\blacksquare (V_{GS} - 1.5)^2 = \frac{7 - V_{GS}}{5}$$



$$\bullet (V_{GS} - 1.5)^2 = \frac{7 - V_{GS}}{5}$$

$$\circ V_{GS}^2 + (1.5)^2 - 2(V_{GS})(1.5) = \frac{7 - V_{GS}}{5}$$

$$\blacksquare V_{GS}^2 + 2.25 - 3V_{GS} = \frac{7 - V_{GS}}{5}$$

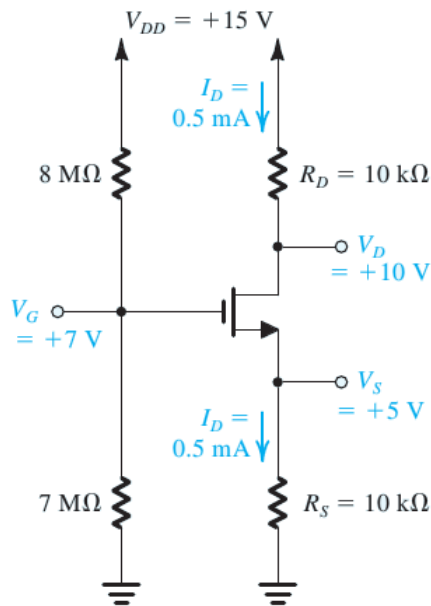
$$\blacksquare 5V_{GS}^2 + 11.25 - 15V_{GS} = 7 - V_{GS}$$

$$\blacksquare 5V_{GS}^2 + 4.25 - 14V_{GS} = 0$$

$$\blacksquare 5V_{GS}^2 - 14V_{GS} + 4.25 = 0$$

$$\blacksquare = \Rightarrow V_{GS} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{14 \pm \sqrt{14^2 - 4(5)(4.25)}}{2(5)}$$

- $V_{GS} = \frac{14 \pm \sqrt{111}}{10} = 2.4536, 0.34643$
- As $V_t = 1.5V$ and as I_D is not zero $\Rightarrow V_{GS}$ can not be equal to 0.34643V
- $\Rightarrow V_{GS} = 2.4536V$



- As $I_D = \frac{7 - V_{GS}}{10k} \Rightarrow I_D = \frac{7 - 2.4536}{10k} = 0.45464mA$
- $I_D = 0.5mA$ when initially $V_t = 1V$
 - and $I_D = 0.45464mA$ when $V_t = 1.5V$
 - $\Rightarrow$
 - the change in I_D is
 - $\Delta I_D = 0.45464m - 0.5m = -0.04536mA$
 - thus the %age change is
 - $\frac{\Delta I_D}{I_D} \times 100 = \frac{-0.04536m}{0.5m} \times 100 = -9.072\%$

Exercise 5.33

- Consider the MOSFET in Example 5.12 when fixed- V_{GS} bias is used. Find the required value of V_{GS} to establish a dc bias current $I_D = 0.5mA$. Recall that the device parameters are $V_t = 1V$, $k_n'(W/L) = 1mA/V^2$, and $\lambda = 0$. What is the percentage change in I_D obtained when the transistor is replaced with another having $V_t = 1.5V$?

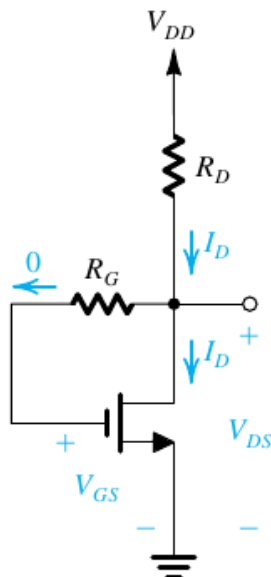
Solution:

- here $I_D = 0.5mA$
 - $I_D = 0.5m = \frac{1}{2}k_n' \frac{W}{L} (V_{GS} - V_t)^2 \Rightarrow 0.5m = \frac{1}{2}(1m)(V_{GS} - V_t)^2$

- $1 = (V_{GS} - V_t)^2 \Rightarrow V_{GS} - V_t = 1$ or $V_{GS} = 1 + 1 = 2V$
- Now if $V_t = 1.5V$
 - $I_D = \frac{1}{2}k_n' \frac{W}{L}(V_{GS} - V_t)^2 = \frac{1}{2}(1m)(V_{GS} - 1.5)^2$
 - As V_{GS} is fixed to 2V
 - $\Rightarrow I_D = \frac{1}{2}(1m)(2 - 1.5)^2 = 0.125mA$
- $\Delta I_D = 0.125m - 0.5m = -0.375mA$
- and $\frac{\Delta I_D}{I_D}$ in percentage = $\frac{-0.375m}{0.5m} \times 100 = -75\%$

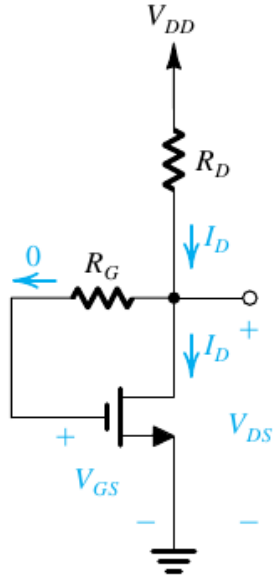
Biasing Using a Drain-to-Gate Feedback Resistor

- Another effective discrete-circuit biasing arrangement is possible by
 - connecting a feedback resistor between the drain and the gate
 - here the resistance R_G (typically in mega ohms range) forces
 - the dc voltage at the gate to be equal to that at the drain ($\because I_G = 0$)
 - $\Rightarrow V_G = V_D$ or $V_{GS} = V_{DS}$
 - by KVL
 - $V_{DD} = I_D R_D + V_{DS}$
 - $V_{DD} = I_D R_D + V_{GS} \because V_{GS} = V_{DS}$
 - or $V_{DD} = V_{GS} + I_D R_D$

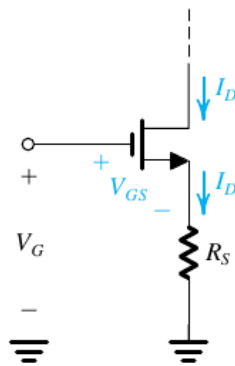


- $V_{DD} = V_{GS} + I_D R_D$
 - which is identical in form to the one obtained for the “biasing by fixing V_G and connecting R_S in the source”

- $V_{DD} = V_{GS} + I_D R_D$



-
- $V_G = V_{GS} + I_D R_S$

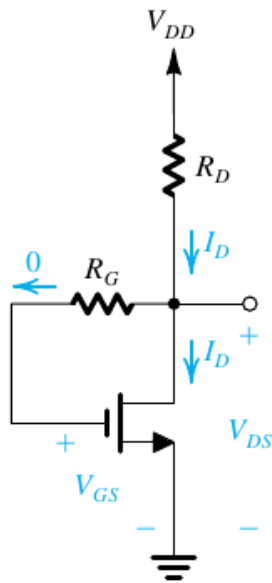


(a)

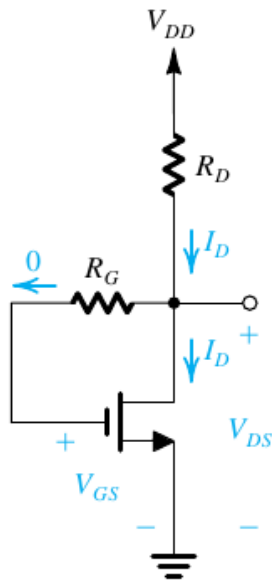
- $V_{DD} = V_{GS} + I_D R_D$

- thus, if I_D for some reason changes, say increases, then

- the above equation indicate that V_{GS} must decrease.
 - This decrease in V_{GS} in turn causes a decrease in I_D
- thus the negative feedback provided by R_G works to keep
 - the value of I_D as constant as possible



- to use this circuit as a CS amplifier, the input voltage
 - signal can be applied to the gate via a coupling capacitor,
 - so as not to disturb the dc bias condition
 - the amplified output signal at the drain can be coupled to another part
 - of the circuit again via a coupling capacitor

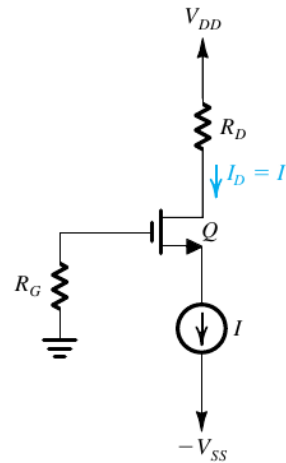


Biasing Using a Constant-Current Source

- Another effective way for biasing a MOSFET is shown in fig and it utilizes a constant-current source
 - here R_G (usually in mega ohms range) establishes a dc ground
 - at the gate and presents a large resistance to an input

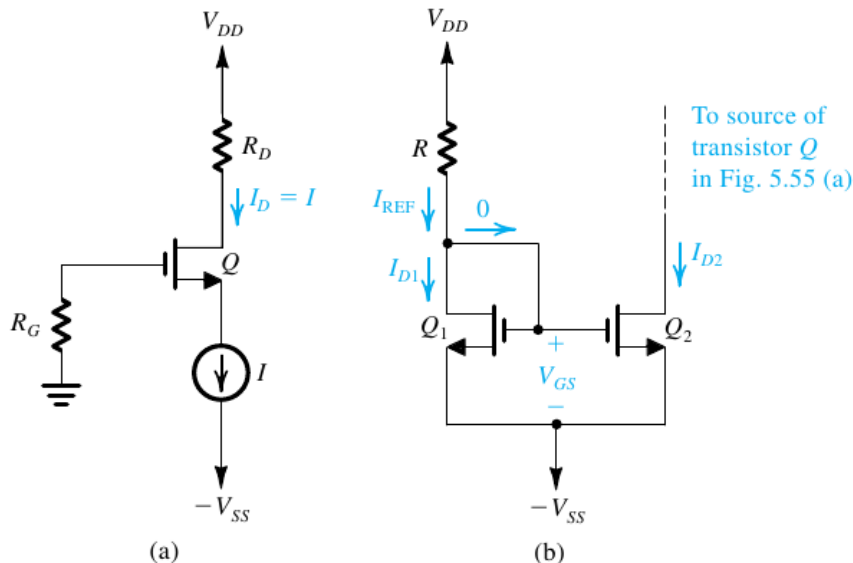
signal source

- that can be coupled to the gate via a coupling capacitor
- resistor R_D establishes an appropriate dc voltage at the drain
 - to allow for the required signal swing while ensuring
 - that the transistor always remains in the saturation region



(a)

- the current source I is implemented by the current mirror circuit shown in fig b



(a)

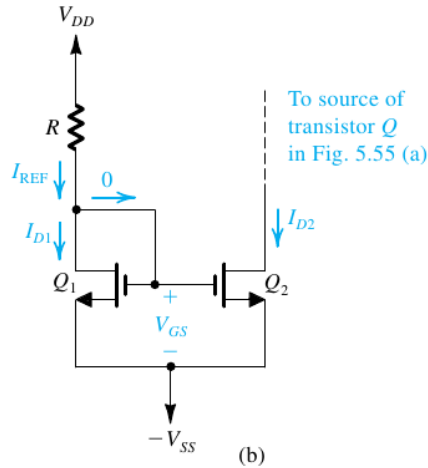
(b)

Figure 5.55 (a) Biasing the MOSFET using a constant-current source I .

- (b) Implementation of the constant-current source I using a current mirror.

- for the current mirror, the drain of Q_1 is shorted to its gate
 - $\Rightarrow Q_1$ is operating in the saturation region

- As for saturation $V_{DS} \geq V_{GS} - V_t$
 - here $V_{DS} = V_{GS} \Rightarrow 0 \geq -V_t$ which is true
- neglecting the channel length modulation i.e. $\lambda = 0$
 - $\Rightarrow I_{D1} = \frac{1}{2}k_n' \left(\frac{W}{L}\right)_1 (V_{GS} - V_t)^2$
- here the drain current I_{D1} is supplied by V_{DD} through R
 - As gate current is zero $\Rightarrow I_{D1} = I_{REF}$

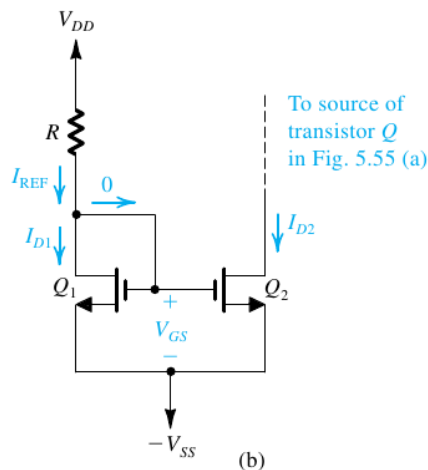


• By KVL

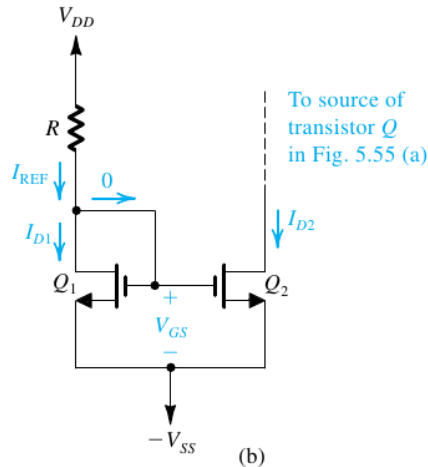
○ $V_{DD} = I_{REF}R + V_{GS} + (-V_{SS})$

- or $(V_{DD} + V_{SS}) = V_{GS} + I_{REF}R$ where $I_{REF} = I_{D1}$
- Note that this equation is of the same form as the one in the previous biasing schemes
 - $\Rightarrow$ the drain current will not change much if Q_1 is replaced by another transistor of same type

■ $I_{REF} = I_{D1} = \frac{(V_{DD} + V_{SS}) - V_{GS}}{R}$



- now the transistor Q_2 has the same V_{GS} as Q_1
 - thus if we assume that Q_2 is in saturation
 - $\Rightarrow I = I_{D2} = \frac{1}{2}k_n' \left(\frac{W}{L}\right)_2 (V_{GS} - V_t)^2$
 - or $I_{D2} / \left(\frac{W}{L}\right)_2 = \frac{1}{2}k_n' (V_{GS} - V_t)^2$
 - as V_{GS} , V_t and k_n' are the same for Q_1 and Q_2 , and as $I_{D1} = \frac{1}{2}k_n' \left(\frac{W}{L}\right)_1 (V_{GS} - V_t)^2$
 - $\Rightarrow I_{D1} / \left(\frac{W}{L}\right)_1 = \frac{1}{2}k_n' (V_{GS} - V_t)^2$



- $\Rightarrow I_{D2} / \left(\frac{W}{L}\right)_2 = \frac{1}{2}k_n' (V_{GS} - V_t)^2 = I_{D1} / \left(\frac{W}{L}\right)_1$
 - or $I_{D2} / \left(\frac{W}{L}\right)_2 = I_{D1} / \left(\frac{W}{L}\right)_1$
- or $I_{D2} / \left(\frac{W}{L}\right)_2 = I_{D1} / \left(\frac{W}{L}\right)_1$
 - As $I_{D2} = I$ and $I_{D1} = I_{REF}$
 - $\Rightarrow I / \left(\frac{W}{L}\right)_2 = I_{REF} / \left(\frac{W}{L}\right)_1$
 - $\Rightarrow$

$$I = I_{REF} \frac{\left(\frac{W}{L}\right)_2}{\left(\frac{W}{L}\right)_1}$$

- thus I is related to I_{REF} by the ratio of the aspect ratios of Q_1 and Q_2
- This circuit is known as a current mirror, and is very popular in the design of IC MOS amplifiers

